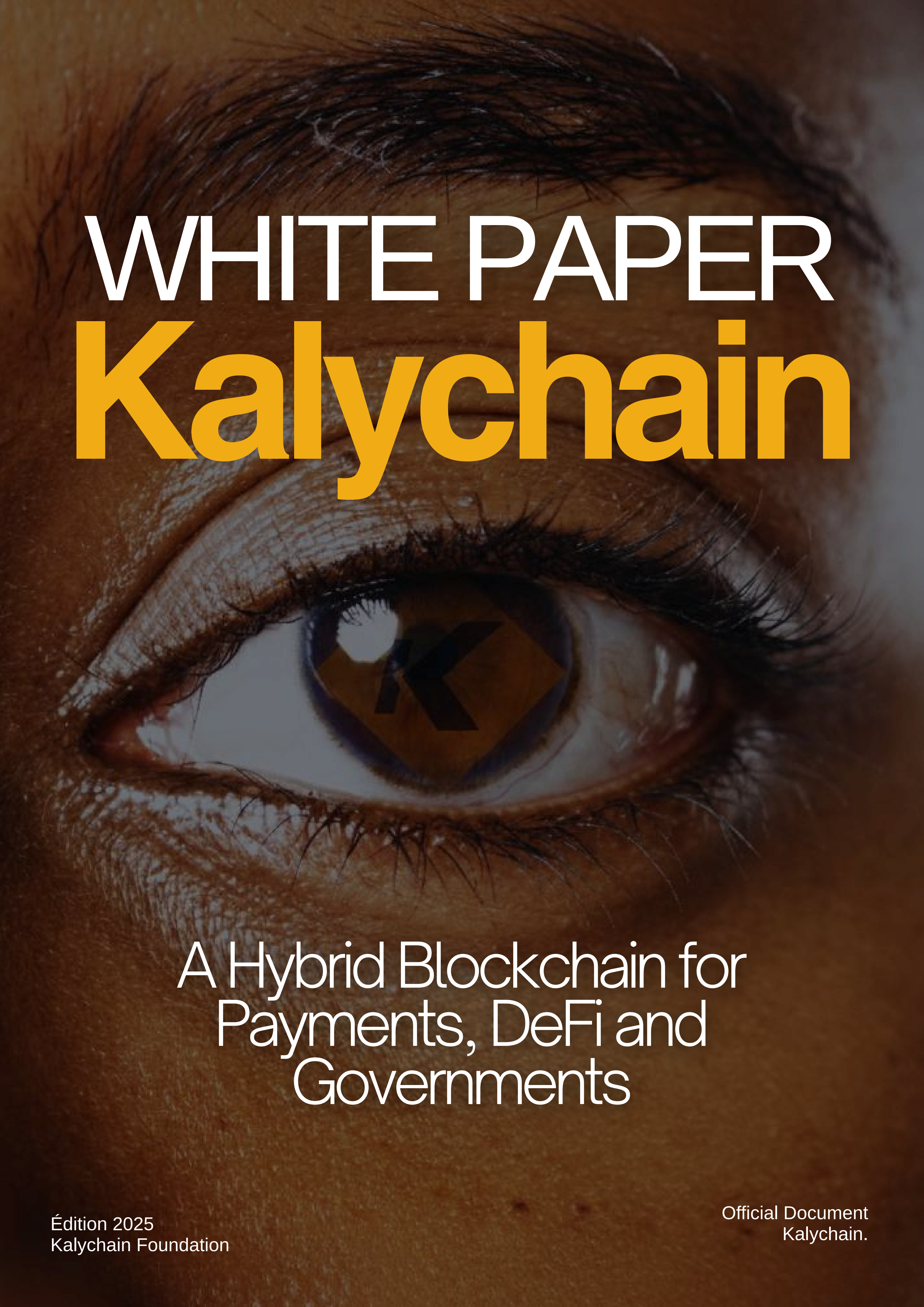

WHITE PAPER
Kalychain

A Hybrid Blockchain for Payments, DeFi and Governments

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Kalychain Foundation

Official Document
Kalychain.

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CHAPITRE 1

Executive Summary

Introduction Kalychain

KalyChain is an EVM-compliant Layer 1 blockchain, designed for real finance: payments, SMEs, public institutions and real-world assets. It combines a robust technical architecture, a hybrid governance Foundation + DAO and native financial primitives (KUSD, KalySwap, KLC) to offer a credible, interoperable and sustainable infrastructure. This document presents the vision, key differentiators, products and principles that guide execution—without any ranking or market cap objectives.

1. Real finance vision

Create the trusted infrastructure that connects traditional finance, challenge and everyday economy:

- For citizens: almost zero-cost stablecoin payments and UX for the general public.
- For SMEs: smart billing, onchain treasury, easy access to liquidity and credit.
- For public institutions: budget transparency, digital identity, G2C payment rails.
- For financial institutions: interoperable rails, compliance and predictable purpose.

Positioning: if Ethereum is the developer platform, BNB the trading ecosystem, and Solana the retail experience, KalyChain specializes in real finance, compliance, and institutional adoption.

2. Key differentiators

- Transfers of stablecoins without gas (gasless)
- Stablecoin transactions (KUSD, and supported stablecoins) can be sponsored by the DAO according to defined policies: thresholds, limits, KYC/usage.
- Effect: elimination of the greatest UX obstacle to the acceptance of online payments; supports micro-payments and G2C on a large scale.
- Hybrid governance Foundation + DAO
- The KalyChain Foundation (non-profit entity) provides legal legitimacy and executes DAO votes.
- The DAO decides on budgets, key parameters, listings, and the roadmap; transparency via a Transparency Hub.
- Full EVM compatibility
- Direct deployment of Ethereum smart contracts, standard tooling (Metamask, Hardhat, The Graph), quick wallet/SDK integrations.
- Consensus trajectory: from QBFT to HotStuff (on client Besu)
- Fast finality, byzantine tolerance, complexity O(n), BLS aggregation.
- Target: 256–1000 validators, purpose 2–5s, sustained flow 400–500 TPS with extended validator sets.

3. Products

- KUSD (stablecoin)
- KUSD Core: 150% overcollateralized (BTC/ETH), transparent mint/redeem process, robust oracles, governed parameters.
- KUSD Plus: extension via stable collaterals/RWAs with regulated partners, emission caps and risk controls.
- KalySwap (native DEX)
- AMM optimized for stable pairs and concentrated liquidity; aggregator integrations; anti-manipulation policy; fee sharing governed.
- KLC (native token)
- Utilities: gas fees, staking (network security), DAO governance, ecosystem value capture (DEX/stablecoin according to regulation), locks/burn/gKLC.

4. Guiding principles

- Security first
- Multi-firm audits on critical components (consensus, bridge, stablecoin, AMM).
- Bug bounty public; OpSec practices (HSM/key, rotation, access), timelocks and rollback plans.
- Pragmatic compliance
- KYC/AML/Travel Rule according to the use cases (notably for institutional flows).
- Clear separation Foundation/DAO/KALYSSI; certificates and proof of reservations for KUSD.
- Real interoperability
- Bridges to major ecosystems (ETH/BNB/Polygon), oracles (Chainlink/Pyth), indexing (The Graph), public analytics (Dune).
- Economic sustainability
- Emissions and fee policies aligned with network security and adoption.
- Gasless under budgetary governance (levels, limits, co-sponsors) to avoid any risk of unsustainability.

5. Why now

The convergence of three dynamics creates an adoption window:

- RWA and institutional payments are looking for programmable rails with finality and compliance.
- Maturity of EVM tools and infrastructure (oracles, indexing, wallets) reduces the cost of integration.
- Need for transactional 'general public' UX (gasless, reliability, low latency) to capture everyday life.

6. What KalyChain changes concretely

- G2C and B2B payments: massive and frequent flows become economically viable onchain thanks to the sponsored gasless and KUSD.
- Onchain treasury for SMEs: stable accounts, instant settlement, compliant reporting, access to liquidity.
- Public transparency: budgets and government transfers traceable in real time, with digital identity and access control.

7. Transparency commitments

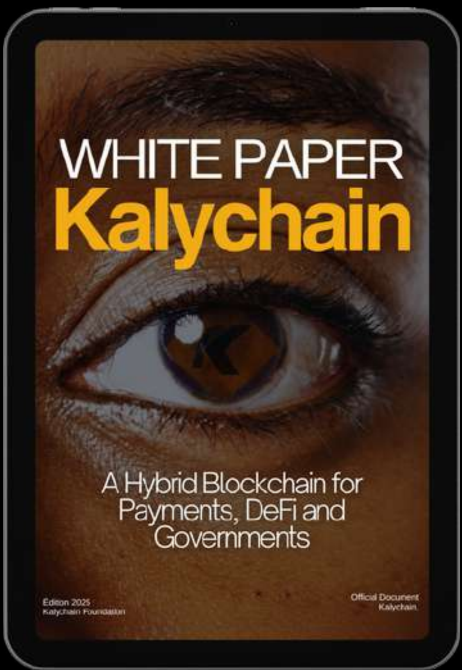
- Transparency Hub public
- Contract addresses and KUSD reserves, network metrics (SLO/SLA), DAO vote histories, audit reports and attestations.

- Verifiable metrics
- Official Dune/The Graph dashboards; publications of benchmark methodologies and known limits.

8. What this white paper covers

- System Architecture: EVM execution, expense policy/MEV, interop, observability.
- Consensus and Performance: QBFT status, HotStuff design on Besu, migration, benchmarks, risks/mitigations.
- KUSD: design, risk framework, compliance, oracles and liquidations.
- Gasless: mechanism, anti-abuse policies, economic model and integrations.
- Stack challenge: KalySwap,gKLC/wKLC, perimeter compliance, product roadmap.
- KLC Tokenomics: roles, emissions/vesting, sinks and sustainability.
- Governance: Foundation + DAO, KIP processes, committees, Transparency Hub.
- Security and Resilience: audits, OpSec, incident plans/rollback.
- Interop and Integrations: bridges, oracles, standards, SDKs, wallets.
- Institutional use cases: G2C/B2B flows, treasury, priority markets, partners.
- Market and Liquidity: DAO playbook based on technical/qualitative criteria (without market cap).
- Financial model: revenues/costs, budgetary policy, sensitivities, reporting Foundation.
- 10-year roadmap: phases, technical deliverables and transition criteria.
- Risks and Legal: risk matrix, assumptions, regulation and jurisdictions.

KalyChain is building payment and institutional finance rails on a secure and interoperable L1 EVM, combining overcollateralized stablecoin (KUSD), managed gasless transfers, disciplined native challenge and foundation + DAO governance, with a consensus trajectory towards HotStuff to support decentralization and purpose tailored to real-world requirements.



CHAPITRE 2

Problem, market landscape and timing

2.1 The points to be addressed

- Costly and slow cross-border payments
- Costs 6–10% for remittances, delays 1–5 days, lack of strong purpose.
- Integration complexity (corresponding banks, cut-off times, chargebacks).
- Frictions for SMEs
- Limited access to credit/cash, high PSP fees, settlement delays D+1 to D+7.
- Fragmented IT infrastructure, fraud and chargeback, multi-system reporting.
- Insufficient public adoption of blockchain rails
- UX of misunderstood gas fees, volatility of fees/latency, multichain fragmentation.
- Lack of purpose and compliance guarantees for government cases.
- Institutional challenge
- Heterogeneous KYC/AML/Travel Rule; auditability and traceability expected.
- Reserves and collateral of non-transparent stablecoins, counterparty risks.

These constraints explain why the majority of flows remain on SWIFT/PSPs, while the challenge remains confined to speculative uses.

2.2 Where the current L1/L2 fail for the "real finance"

- Bitcoin
  - Strength: credibility 'value reserve'.
  - Limits: latency/fees, limited scripting for large-scale programmable payments.

- Ethereum
  - Strength: ecosystem, security, standards.
  - Limits: variable fees, congestion; adapted to complex assets, less to mass micro-payments.

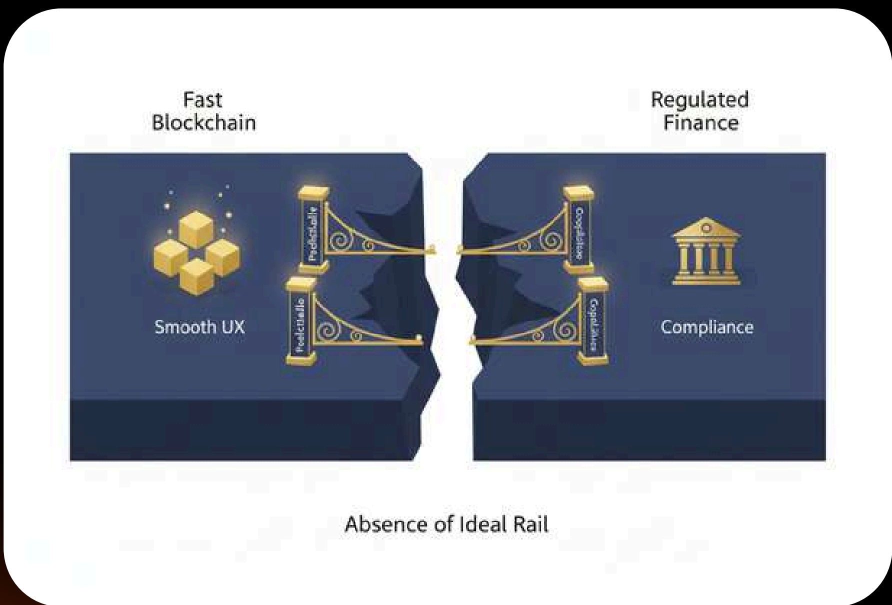
- BNB Chain
  - Strength: low costs, wide distribution.
  - Limits: perception of centralization; difficulty of acceptance by public institutions.

- Solana
  - Strength: performance, consumer UX.
  - Limits: historical failures, governance/perceived reliability trajectory for state requirements.

- L2 (Optimistic/ZK rollups)
  - Strength: reduced fees, EVM compat.
  - Limits: bridge complexity, purpose delays (withdrawals), L1 dependency for perceived security.

- Existing stablecoins
  - Force : adoption large (USDT/USDC).
  - Limits: transparency of reserves unequal; very few "gasless" UX; few integrated rails for SMEs/States.

Conclusion: No rail today combines UX 'gasless', predictable purpose, pragmatic compliance, credible governance and EVM tooling, specially oriented payments/SMEs/States.



2.3 KalyChain Positioning

- Specialization: real finance (payments, treasury, G2C) rather than speculative trading.
- Structuring differentiators:
- Gasless stablecoins sponsored by DAO (policies, limits, compliance).
- Hybrid governance Foundation + DAO (institutional execution + on-chain transparency).
- QBFT consensus trajectory HotStuff (256–1000 validators, purpose 2–5 s).
- Overcollateralized KUSD (Core 150% BTC/ETH) + risk framework and certificates.
- Integration philosophy:
- Complete EVM compat (zero friction for dApps/tools).
- Prioritized interop with the ETH/BNB/Polygon ecosystem; oracles Chainlink/Pyth; indexation The Graph; analytics Dune.
- Transparency Hub as an anchor point for trust (audits, reserves, network metrics).

2.4 Why now (Timing)

- RWA convergence x compliance x UX
- Institutions are looking for programmable rails with finality/auditability.
- The regulation is structured (licenses, CASP, Travel Rule), making a 'over-politicized' stablecoin viable at risk.
- The 'gasless' UX lowers the main barrier of retail and G2C adoption.
- Technical maturity
- Standardized EVM tools; more robust oracles/bridges; best security and observability practices available.
- Progress of modern BFT (HotStuff) to move from sets of 30 validators to 256–1000 with fast finalization.
- Macro request
- Increase in PSP costs and geopolitical tensions on the current tracks.
- Need for States to improve the transparency and effectiveness of social transfers.
- Growing crypto-native SMEs, looking for stable cash flow and instant settlements.

2.5 Iconic Usage Scenarios

- G2C on a large scale
- Distribution of social aid in KUSD, without gas friction, with certificates of reserves and public journalization.
- Cross-border B2B payments
- Billing/matching on KalyChain, KUSD gasless settlement; auditable reporting for external audit.
- Cash SME onchain
- KUSD accounts, maturity/supplier management; access to credit via simple collateralization; reporting exportable.
- RWA and institutional funds
- Coffres tokenized via regulated partners; integration oracles ; governance of risk parameters by dedicated committees.

2.6 Measurement principles (without ranking objectives)

- Reliability and purpose
- Network SLOs: availability, purpose latency, TX failure rate.
- Integrity: incidents, resolution time, post-mortem transparency.
- Useful adoption
- KUSD payment volumes, merchant accounts, PSP/bank integrations.
- Number of KYC/KYB opt-in entities for institutional flows.
- Economic health
- DEX/CEX liquidity depth on stable pairs and KLC.
- KUSD coverage ratio, composition of the bond, published stress tests.

These metrics feed the Transparency Hub, without referring to market cap or rankings.

The market needs programmable payment rails with purpose, compliance and viable UX for everyday life. KalyChain responds with a real finance specialized L1 EVM: sponsored and governed gasless, overcollateralized stablecoin with risk framework, Foundation+DAO governance, and a consensus trajectory towards HotStuff for decentralization and institutional performance.

CHAPITRE 3

System Architecture

Give a clear, complete and verifiable view of KalyChain's L1 EVM architecture, fee/MEV policies, interoperability (bridges, oracles, indexing) and observability (SLOs, monitoring, public metrics). This section serves as a basis for technical teams, auditors, and integrators (PSP, banks, dApps).

3.1 Overall logical view

Layers and components:

Network and consensus

- Validator nodes (staking KLC), client EVM Besu, current consensus QBFT and HotStuff trajectory (see Chap. 4).
- P2P Gossip, state and block synchronization, certificate/signature propagation.

EVM Execution

- Moteur EVM compatible Solidity/Vyper, opcodes EVM standard, gas accounting, state trie (Merkle/Patricia).
- Mechanism of refunds, usual precompiled (ECRecover, SHA256, BN256).

Mempool and scheduling

- Mempool prioritized by actual fees; separate gasless queue via sponsored relayers (policy engine).
- Inclusion policy: limits per block, starvation prevention, anti-spam constraints.

Fee market

- Dynamically adjusted block base fee (model EIP-1559-like adapted); priority tips.
- DAO grants for eligible gasless transactions (stablecoins): support via relayers and smart sponsor.

MEV and equity policy

- Scheduling rules: priority by fees, MEV "bundle" limits, no private owner mempool by default.
- Commitment to transparency: periodic publication of detected MEV metrics and mitigation policies.

System services

- Oracles, gasless relayers, indexing The Graph, public/private RPCs, archives.
- Official bridge(s) with limits by asset ('caps') and monitoring.

External interfaces:

- Compatible JSON-RPC RPC (eth_, net_, web3_), HTTP(s)/WebSocket endpoints, rate limits and SLA.
- SDKs et outils : Hardhat/Foundry, Ethers/Web3.js, wallets (Metamask, Ledger, OKX), APIs d'indexation.

3.2 Execution L1 EVM

- Full EVM compatibility:
- ABI standard, events/logs, topics, filtres, traces ; compatibility The Graph and Dune.
- Deployment of contracts via standard toolchains; support of the main ERCs (20/721/1155/2612).
- Gas parameters and limits:
- Gas target per block calibrated to support 400–500 TPS in stable payments/DEX profiles.
- Policy of gradual increase in the gas target subject to DAO voting after performance/network audits.
- Precompiled and cryptography:
- Precompiled ECDSA/BN256 standards; BLS for consensus HotStuff (client side/consensus, cf. Chap. 4).
- Curves, key sizes, and parameters documented in the annexes.

3.3 Mempool and scheduling policy

- Queues:
- Standard mempool: tri par effective gas price = base fee + tip.
- File gasless: sponsored stablecoin transactions validated by the policy engine; inserted with defined priority and quota per block.
- Anti-spam and anti-abuse:
- Rate per address/IP for public RPC; minimum deposit for relayers; sanctions lists applied at the sponsor level.
- Limits per block for gasless (e.g. X% of the gas per block) in order to preserve the network liveness.
- Equitable inclusion:
- Starvation prevention: reserved slots for non-gasless tx and eligible gasless tx.
- Public logging: inclusion metrics, mempool latency block.

3.4 Fee Market and Gasless

- Basic model:
- Dynamically adjusted base fee (block fill target) + tips for priority.
- Collected fees partially burned and/or redirected according to KLC tokenomics (cf. Chap. 8).
- Gasless subsidies:
- Smart contract sponsor controlled by the DAO; periodic budgets; eligibility rules (amount, frequency, contextual KYC/KYB).
- Authorized relayers (allowlist) with log obligations and proof of execution.
- Governance and transparency:
- KIP proposals to modify quotas, eligibility, budget ceilings.
- Public dashboard: gasless cost, aggregated beneficiaries, impact on adoption.

3.5 SRM: policy and mitigation

- Targeted SRM sources:
- tx priority, sandwiching on DEX, mempool reordering.
- Policy:
- No private owner mempool by default; equality of access to public RPC.
- Encouragement of 'fair ordering' techniques (lottery-based, time-based) on canary testnets before possible activation.
- Observability:
- Detection of MEV patterns on DEX; publication of monthly reports; mitigation proposals to be submitted to the DAO.

3.6 Interoperability

- Official Bridges:
- Conception : bridge(s) multi-sig/validator light clients selon actif ; explicit trust assumptions.
- Caps by asset and by period; oracles of price to manage flow limits; 24/7 monitoring.
- Incident process: bridge emergency pause, restitution plans and public notifications.
- Oracles:
- Planned integrations: Chainlink, Pyth for price BTC/ETH/stables and RWA baskets.
- Redundancy policy: fallback providers, report hashing, bypass thresholds, max latency.
- Indexation and data:
- The Graph official subgraphs for KUSD, KalySwap, governance.
- Real-time export to Dune/Flipside; published event schedules.

3.7 Observability, SLOs and Monitoring

- SLOs network proposed:
- Public RPC availability: 99.9% monthly.
- Finality latency: median 3 s; p95 ≤ 5 s (post-HotStuff).
- Transaction failure rate: 0.5% excluding application errors.
- Telemetry and alerting:
- Metrics: block rate, gas used, mempool backlog, orphan rate, reorgs, P2P latencies, RPC errors.
- Logs signed for relayers gasless; traceability of subsidies; budget alerts.
- Public transparency:
- Official dashboards (Dune/The Graph) with documented methodologies.
- Post-mortem reports published within 7 days after a major incident.

3.8 Environments and deployments

- Networks:
- Mainnet, Testnet (pre-production), Canary (experimental features: fair ordering, new gasless quotas).
- Clients and versions:
- Main client Besu with consensus modules; semantic versioning ; upgrade schedule (hard forks) with timelock and DAO vote.
- CPP and infrastructure:
- public RPC managed by the Foundation + third-party operators; rate limits; endpoints d'archive.
- Hardware recommendations for validators and indexing operators.

3.9 Architecture Security (overview)

- Access controls and keys:
- Separation of roles: RPC operators, validators, bridge signers, DAO guardians; HSM for sensitive keys.
- Secure upgrades:
- Timelocks, multi-sig, "pause" on sensitive modules (bridge, gasless sponsor).
- Defense in depth:
- Throughput limits, strict validation of RPC inputs, sandboxing of relayers, regular audits of configurations.

The KalyChain architecture combines a standard EVM execution, a transparent fee market with managed gasless support, a balanced MEV policy, prudent interoperability (bridges/oracles with caps and monitoring) and 'proof-first' observability via public SLOs. This database prepares the transition to HotStuff (fast finality, more validators) and supports real finance use cases.

CHAPITRE 4

Consensus and performance : From QBFT to HOTSTUFF

Explain the current state (PoS + QBFT), detail the HotStuff design on Besu, define the integration/migration plan, document the benchmark methodology and establish a risk/mitigation matrix. This section should allow auditors, integrators and validator operators to evaluate the security, purpose and scalability of KalyChain.

4.1 Current Status PoS + QBFT

- Security model
- BFT allowed by set of validators staked in KLC.
- Byzantine fault tolerance: security ensured if $f < n/3$ validators are corrupted.
- Operational parameters (indicative)
- Set size: 64–128 active validators.
- Block time: 2–3 s; practical purpose in 2–3 blocks (6–9 s).
- Leader rotation: deterministic by period/block height.
- Observed limits
- Communication overheads $O(n\ 2)$ in commit latency phases increasing beyond ~128–160 validators.
- Increased sensitivity to network latency peak/leader failures.
- Aggregation of signatures limited in terms of performance (ECDSA aggregated outside the chain more complex).
- Guarantees
- Strict purpose (without deep forks) in the absence of Byzantines n/3.
- Proven EVM Besu compatibility; tooling mature.

Conclusion: QBFT is suitable for a moderate set of validators and a finality latency in the order of 6–9 s. To support expanded decentralization (256–1000 validators) and a 2–5 s finality, we migrate to HotStuff.

4.2 Design HotStuff on Besu

- HotStuff Principles
- 3-phase BFT Protocol with quorum certificates (QC): Prepare Pre-Contract Commit.
- O(n) messaging thanks to QCs and signature aggregation (BLS).
- Liveness reactive: finality latency tracks network latency under a correct leader.
- Safety: as long as they are < n/3 byzantine, it is impossible to finalize two conflicting blocks.
- Implementation architecture
- Client: native Java implementation within Besu (HotStuff consensus module).
- Cryptography: BLS12-381 for signature aggregation; BLS keys separated from the execution ECDSA keys.
- Leader rotation: weighted round-robin (stake, performance), with quick view change in case of a faulty leader.
- Pipelining: possibility of Prepare/Pre-Commit pipeliner for stable high throughput.
- Target parameters
- Set of validators: 256 to 1000.
- Target block time: 1–2 s.
- Purpose: median 2–3 s; p95 4–5 s under normal network conditions.
- Supported debit: 400–500 TPS on stable payments/DEX workloads, with 256+ validators.
- Management of keys and signatures
- Participation scheme: on-chain registers of BLS keys; change of keys with timelock.
- Aggregation: aggregators designated per round (can be the leader) with automatic fallback.
- Fault tolerance and synchronization
- View-change: timeouts adaptatifs; quick re-election to maintain liveness.
- Synchronization: recovery by state and QCs; prévention des “locking” deadlocks via rules HotStuff.

4.3 Integration and Migration

- Phasage
- Testnet-HS: deployment of a dedicated HotStuff test network, heavy instrumentation (latency, loss, churn).
- Dual-run on main Testnet: QBFT (production) + HotStuff (mirror) with real load replay (shadow traffic).
- Canary Mainnet: subset of validators running HotStuff in parallel; checkpoints and QCs compared off-chain.
- Activation Mainnet: switch by upgrade planned via DAO, timelock 7–14 days, multi-sig guardian key.
- Security of upgrades
- Timelocks and DAO voting for activation; publication of a rollback plan in advance.
- Guardian keys: ability to 'pause' the HotStuff/bridge module in case of a critical incident, with transparency.
- EVM compatibility and tooling
- No change at the opcode/ABI level; transparency for the dApps.
- Update of validator clients; orchestration scripts and playbook operators.

4.4 Benchmarks and Methodology

- Load profiles
- Payments: simple tx ERC-20 (KUSD) with gasless transfers included.
- DEX stable: swaps via AMM stable; bursts during oracle updates.
- Production mix: 70% payments, 25% DEX, 5% governance/misc.
- Test environment
- Topology: 256/512/1000 geographically distributed validators (8 regions).
- Network: RTT latency 50–150 ms, loss 0.1–0.5%, jitter; scenarios of temporary partitions.
- Matériel: CPU 8–16 vCPU, RAM 32–64 GB, SSD NVMe; HSM or enclaves for BLS keys if available.
- Measures
- Purpose latency (median, p95, p99), throughput (confirmed TPS), reconfig/view-change rate, orphan rate.
- Coûts CPU/RAM/disk IOPS; overhead d'agrégation BLS; impact mempool gasless.
- Procedure
- Warm-up 30 min; palier 1 → N (ramp-up); stability bracket 60–120 min; ramp-down.
- Public reports: scripts, configs, runbooks, known limits; reproducibility by third parties.

4.5 Risks and Mitigations

- Failure/abuse of the leader
- Risk: Byzantine leader slows the finality.
- Mitigation: adaptive timeouts, fast rotation, stake penalties for non-participation.
- Network partitions and extreme latency
- Risk: repeated view-changes, throughput decrease.
- Mitigation: timeout settings by region; QCs caches; active monitoring; RPC failover rules.
- BLS/aggregator synchronization
- Risk: aggregator at fault; partial signatures.
- Mitigation: backup aggregators; redundancy scheme; load tests on aggregation.
- DoS on consensus
- Risk: P2P overload, flooding attacks.
- Mitigation: rate limits P2P, scoring pairs, banning temporaire; resources reserved consensus vs RPC.
- Extensions of the validator set
- Risk: frequent churn, view-change instability.
- Mitigation: input/output windows, churn caps, reconfiguration epochs.
- Besu implementation
- Risk: protocol/consensus bugs.
- Mitigation: audits indépendants, fuzzing réseau, chaos testing; canary extended before activation.

4.6 Operation and Consensus Governance

- Committees and responsibilities
- Core Protocol committee: propose versions/parameters; publishes test reports.
- Security Committee: valid audits/fuzzing; maintains playbooks incident.
- Governed settings (via KIP)
- Target block time, timeouts, leader rotation strategy, BLS thresholds, max validators size by epoch.
- Transparency
- Publication of binaries, checksums, upgrade guides, and benchmark results.
- Logging of consensus and post-mortem incidents in 7 days.

KalyChain is evolving from a QBFT suitable for medium sets to HotStuff on Besu to support 256–1000 validators with finality 2–5 s and 400–500 TPS on payout/DEX charges. The migration is phased (testnet dual-run canary mainnet), framed by guardrails (timelocks, guardians, rollback) and validated by public benchmarks and independent audits, in order to offer a purpose and decentralization adapted to real finance.

CHAPITRE 5

KUSD : design, Risk and et Compliance

Define precisely the architecture of KUSD (native stablecoin), its collateralization and stability mechanisms, the governance of risk parameters, as well as the compliance and attestation framework. The goal is to provide a verifiable, auditable and sustainable design for institutional uses, SMEs and consumer payments.

5.1 Overview and objectives

- Role of KUSD:
- Unit of account and priority settlement for payments, SME treasury and G2C rails.
- Transparently collateralized, on-chain governed, designed for PSP/bank interoperability and integration.
- Targeted properties:
- Parity to the dollar with minimum volatility, high liquidity DEX/CEX, fast finality via L1.
- Resilience in stress test, reliable redemptions, public auditability.

5.2 Architecture of the stablecoin

- Main contracts:
- KUSDToken (ERC-20 + EIP-2612 permit): emission/burn controlled by Core/Plus modules.
- KUSDCoreVault: management of crypto assets (BTC/ETH) and positions, margin calls, liquidations.
- KUSDPlusVault: stable/RWA collateral management via regulated partners, with caps and certificates.
- OracleManager: price flow aggregation (Chainlink/Pyth, fallback), deflection thresholds and max.
- RiskGovernor: storage of parameters (LTV, fees, caps), execution of DAO votes (timelocks).
- RedemptionRouter: redemption request queue, processing window, priorities and fees.
- Accounting:
- Overcollateralized at any time for the Core module.
- Separate accounts, event logs (mint/burn/adjust/liq), reports exportable to subgraphs.

5.3 Collat Core: BTC/ETH at 150% (overcollateralized)

- Initial parameters (examples, governables):
- Max LTV: 66% (minimum ratio of 150%).
- Liquidation threshold: 70–75% LTV, liquidation penalty 5–8%.
- Stability fee (interest rate): 1–3% annual, adjustable by DAO according to demand and costs.
- Mechanics:
- Repository of BTC/ETH tokenized via official bridges and/or wBTC/wETH according to integration.
- Mint KUSD up to LTV max; automatic margin call in case of a fall in headcount.
- Liquidation via dedicated auctions or AMM, with oracles and impact limits.
- Oracles management:
- Median multi-supplier price; deviation threshold (ex: 0.5–1%) and latency (ex: 60s).
- Break mechanism/degraded mode if oracles out of range, validated by multisig guardian with transparency.

5.4 Collat Plus: stable and RWAs via regulated partners

- Eligible items:
- Stablecoins fiat-backed (USDC/USDT) with identified counterparty risk and caps.
- Deposits/short term cash and T-Bills via SPV/regulated partner trusts.
- Governance by caps:
- KUSD Plus global cap (ex. 30–50% of the outstanding amount) and caps by bond/issuer type.
- Mechanism of 'circuit breaker' if certificate missing/delayed or price anomaly.
- Certificates and audits:
- Regular (monthly/quarterly) evidence of underlying assets, signed by third-party auditors.
- Publication on the Transparency Hub; links to reports and notarized evidence.

5.5 Mint, redeem and redemption line

- Mint:
- On Core: permissionless under LTV constraints; possible mint fees (0–10 bps) to manage the request.
- On More: according to availability of the caps and KYC/KYB status if required by the partner.
- Redeem:
- FIFO file with defined operational deadlines; dynamic redemption fees according to liquidity pressure (0–30 bps).
- In stress, activation of 'redemption windows' and temporary haircuts on Plus if the partner imposes deadlines.
- User protection:
- Priority for redeemings of 'small amounts' for daily payments.
- Anti-abuse safeguards (split/merge requests, sybil) and daily limits by address.

5.6 Risk framework

- Governed parameters (via KIP, timelock):
- LTV cibles, liquidation penalties, stability fee, emission caps (Core, Plus, total), oracle deviation/latency, redemption fees/windows.
- Risk committees:
- Risk Committee (multisig + external experts) proposes; Valid DAO.
- Quarterly reports: stress tests, scripted VaR, BTC/ETH/stables correlation analysis, concentration by issuer.
- Stress tests:
- Shocks of -30/-50% BTC/ETH, stable payments (1–3%), suspension of redemption on the partner side.
- Simulations of liquidations and market impact; buyability and normalization time.
- Key indicators:
- Aggregate and per-module collateralization ratio; % in Core vs Plus; maturities of the RWAs; liquidity coverage.

5.7 Liquidations and stability

- Triggers:
- LTV exceeding the threshold or oracles reporting a price out of threshold.
- Execution:
- Auction AMM: tiered auctions or direct liquidation on KalySwap dedicated pools with low slippage.
- Backstops: approved market makers and 'keeper' system under KLC/KUSD incentive.
- Anti-cascade:
- Liquidation caps by block; dynamic braking; coordination with oracles to avoid pro-cyclic spirals.

5.8 Integration of oracles and redundancy

- Multi-oracle:
- Agrégation Chainlink/Pyth + fallback; cross-validation and minimum quorum.
- Monitoring:
- Anomaly detection (outliers, latency, inter-feed divergence).
- Public logging of feed toggles and emergency breaks.

5.9 Compliance: KYC/AML/Travel Rule (depending on use case)

Principle:

- KUSD Core: permissionless for basic ownership/use, under network policies.
- Flux institutionnels: canaux “opt-in compliance” avec KYC/KYB, Travel Rule via partenaires TRISA/Travel Rule providers.
- Lists and sanctions:
- Compliance with sanctions lists at the level of sponsor (gasless) entities and Plus partners.
- Limited and transparent targeted freeze capacities for assets in dispute via legal order.
- Journalisation:
- Logs kept by partners and sponsored relays; sampling accessible to listeners under NDA.

5.10 Transparency and certificates

- Transparency Hub:
- Detailed reserves (Core/Plus), ratios, caps, history of parameter votes and redemption.
- Links to contract audit reports, partner certificates, real-time subgraphs.
- On-chain evidence:
- Chest addresses, mint/burn/redemption events; publication of ABIs.
- Cryptographic “proof of reserves” programs when possible.

5.11 Liquidity and market integrations

- DEX:
- Deep stable pools KUSD/USDC/USDT; KUSD/KLC; incentives via gKLC/wKLC and controlled segments.
- Oracle of stable pools for controlled liquidations.
- CEX and PSP:
- Progressive listing according to technical/qualitative criteria; merchant/PSP integrations for KUSD settlements.
- Interop:
- Bridges avec caps; bridged versions of KUSD on major EVMs for acceptance network.

5.12 Incident Governance and Contingency Plans

- Degraded modes:
- Mint pause on Core/Plus; temporary increase of stability fee/redemption fee; decrease LTV; enlargement redemption windows.
- Roles:
- Guardian multisig with limited mandate, ex post control by DAO (public report).
- Communication:
- Public alerts, incident timelines, post-mortem 7 days.

5.13 Economic model and sustainability

- Revenues:
- Stability fees, redemption/mint fees, possible share of stable AMM fees (according to the legal framework), interest on Plus assets via partners.
- Costs:
- Oracles, audits, operations, gasless subsidies related to KUSD transfers.
- Budgetary policy:
- Quarterly DAO budgets, stress redemption buffer, safety funds.
- Sensitivities:
- Volume payments, TVL collat, market spreads, oracle costs; base/bear/bull scenarios.

KUSD is an overcollateralized and governed stablecoin, articulated in two modules: Core (BTC/ETH 150%) for permissive robustness, and Plus (stable/RWAs) via regulated partners under caps and certificates. Oracles orchestration, governance of risk parameters, transparent mint/redeem/liq mechanisms, and a pragmatic compliance framework make KUSD suitable for payments, to SME treasury and institutional flows while natively integrating with the gasless strategy and KalySwap.

CHAPITRE 6

Transfers of stablecoins without Gas (GASLESS)

Formalise the gasless mechanism for stablecoin transfers (KUSD first), define eligibility policies, quotas and budgets, anti-abuse controls, as well as integration experience for wallets/PSP. This section specifies the security guarantees, financial sustainability and transparency of the program.

6.1 Principles and objectives

- Principle: the DAO sponsors all or part of the gas fees for transfers of stablecoins in accordance with a governed public policy (eligibility, ceilings, quotas).
- Objectives:
- Consumer UX: remove the barrier of gas fees for everyday payments.
- Financial inclusion: making micro-payments, G2C, remittances and frequent B2B more viable.
- Measurement and transparency: costs, volumes, aggregate beneficiaries published on the Transparency Hub.

6.2 Architecture of the gasless mechanism

- Actors
- User issuer and recipient (EOA/contracts).
- Authorized relayers (allowlist) who submit on-chain transactions.
- Smart Sponsor (DAO contract) who reimburses or supplies the relayers according to the policy.
- Policy Engine (contract + optional off-chain verification service) which determines eligibility.
- Flow of a gasless transaction
- The user signs a metatransaction (EIP-712) describing a KUSD transfer.
- The relay verifies eligibility (Policy Engine: amounts, frequency, KYC/KYB if required).
- The relay submits the transaction to the network and pays for the gas.
- The Smart Sponsor reimburses the relay if the transaction complies with the policy and quotas.
- Separation of the files
- Standard mempool file for classic tx.
- Prioritized gasless file with block quotas defined by governance (see 6.5).

6.3 Policy of eligibility (governance policy)

- Basic criteria
- Assets: KUSD priority; possibly other stablecoins whitelists by the DAO.
- Unit amount: limit per transaction (e.g. 500 KUSD).
- Frequency: daily/monthly limits by address (e.g.. 20 tx/day; 300 tx/month).
- Destinations: exclusions of non-compliant contracts (mixers, sanctions lists).
- User segments (examples)
- General public: standard limits (caps/tx + daily quotas).
- Certified Merchants/PSPs: higher caps, under KYB, with reporting obligations.
- G2C/ONG: specific channels approved by DAO with dedicated budgets.
- Contextual compliance
- KYC/KYB opt-in for institutional segments; Travel Rule for regulated transfers via partners.
- Georestrictions if required by the partner's jurisdiction (e.g., sanctions).
- Governance
- All parameters (caps, frequency, exclusion lists, segments) are modified via KIP with timelock and public justification.

6.4 Anti-abuse and security controls

- Anti-sybil and 'farming' of subsidies
- Address clustering heuristics (off-chain), crossed limits by device/fingerprint (for partner relays).
- Symbolic fees or minimal staking on the relay side to discourage spam.
- Rate limits multi-niveaux
- By address, by relay, by block, by minute/hour (burst control).
- Separate quotas for segments (general public vs merchants vs G2C).
- Sanctions lists and blocklists
- Integration of public lists (OFAC, EU, UN) on the relay/sponsor side.
- Minimal on-chain 'deny' mechanism (marked contracts) to prevent reimbursement.
- Observability and limited reversibility
- On-chain detailed logs: hash/meta, relaying, segment, gas cost, policy decision.
- In case of flagrant abuse: temporary suspension from a relay (pause), investigation, public report; no retroactive confiscation of user funds.

6.5 Block quotas and capacity management

- Proposed quotas
- Share of gas per block allocated to the gasless: e.g., 10–25% with absolute ceiling.
- Reserved slots: at least N non-gasless tx per block to avoid starvation.
- Dynamic adjustments
- Throttling algorithm based on gasless backlog, average gas cost and remaining budget.
- Monthly review via DAO according to load and purpose SLO.
- specific SLOs
- Target latency mempool block for tx gasless: median 5 s; p95 10 s.
- Valid refund rate 99.5% for tx eligible.

6.6 Budgets, costs and economic model

- DAO Budget
- Quarterly allocation of the Smart Sponsor (in KLC/KUSD) with alert threshold (e.g. 30% remaining).
- Revisions based on KUSD volumes, actual gas costs, and partner returns.
- Costs and arbitrations
- Average gas cost per transfer (ex. 21k–40k gas on L1 EVM)base fee.
- Co-sponsoring policy: PSP/banks can co-finance for their clients (matching).
- Efficiency mechanisms
- Lots/Batching for massive G2C payments.
- Aggressive gas optimizations of KUSD/transfer contracts.
- Use of off-peak periods with automatic quota adjustment.
- Financial transparency
- Public dashboard: cumulative cost, average/tx cost, top segments, number of active beneficiaries, correlation with adoption.

6.7 UX and integrations (wallets, PSPs, APIs)

- Wallets
- EIP-712 clear prompts (“Kaly DAO sponsored KUSD transfer—no gas to pay”).
- Real-time eligibility indicator; fallback in the 'pay your gas' mode if not eligible.
- PSPs/Merchants
- server SDKs and e-commerce plugins (WooCommerce/Shopify-like via partners).
- Automatic reconciliation: invoice ID, refund logs, accounting export.
- APIs
- Endpoint REST/GraphQL for eligibility estimation, remaining limits tracking and meta-tx submission.
- Webhooks for refund statuses and quota alerts.
- Multi-currency support
- UX conversion: amount in native fiat with oracle FX; acceptance KUSD by default.

6.8 Governance and compliance

- Operational governance
- Committee 'Payments & Gasless' proposes: quotas, budgets, segments; monthly KPIs.
- DAO vote, Foundation execute; quarterly public reports.
- Compliance
- Due diligence process for partner relays; logs and retention compliant.
- Channels “opt-in compliance” for institutions with travel rule requirements.

6.9 Scenarios and examples

- G2C mass payout
  - 100k beneficiaries, 3 payments/month: batching, extended quotas, government co-sponsorship.
- Cross-border merchant
  - 5k payments/day: KYB segment, higher cap/tx, automated reporting.
- Remittances
  - Partner portals that absorb gas for first reception (soft onboarding).

6.10 Impact measurement and safeguards

- KPI d’impact
- Adoption rate (gasless tx/total KUSD tx), average cost/tx, average basket, merchant retention, finality deadlines.
- Guardrails
- Emergency stop: pause of the Smart Sponsor without blocking the classic paid transfers.
- Budget floor/ceilings, policy reviews, regular audit of the sponsor/relay code.
- KPI d’impact
- Adoption rate (gasless tx/total KUSD tx), average cost/tx, average basket, merchant retention, finality deadlines.
- Guardrails
- Emergency stop: pause of the Smart Sponsor without blocking the classic paid transfers.
- Budget floor/ceilings, policy reviews, regular audit of the sponsor/relay code.

KalyChain’s gasless program provides a "consumer" payment UX while remaining safe, governed, and sustainable: transparent eligibility policies, block quotas, DAO budget with co-sponsorship, anti-abuse controls, and wallet/PSP integrations. It maximizes KUSD payment adoption without compromising network liveness or financial discipline.

CHAPITRE 7

KalySwap and challenge Stack

Define KalySwap's functional and economic architecture (native DEX), its price and liquidity formation mechanisms (stable pools, concentrated liquidity), the anti-manipulation/MEV policy, integration with gKLC/wKLC program, as well as the product roadmap and perimeter compliance requirements.

7.1 Role of KalySwap in the ecosystem

- Native liquidity point for KUSD, KLC and bridged assets.
- Price infrastructure for internal oracles (TWAP) and liquidations KUSD.
- Ecosystem incentive channel (gKLC/wKLC) oriented stability of stable pairs and logbook depth.

7.2 AMM Architecture

- Types of pools
- Pools stables (courbe de type stableswap): KUSD/USDC, KUSD/USDT, KUSD/fiat-rwa.
- Concentrated liquidity (CL/MM) for volatile pairs: KLC/USDC, ETH/KUSD, BTC/KUSD.
- Specialized pools "oracle-aware" for KUSD liquidations/auctions with controlled slippage.
- Formulas and parameters
- Stable: invariant adapted (amplification A) to minimize slippage around the peg; Governed.
- CL/MM: discrete ticks, management of positions by price interval; multiple fees (third parties) according to volatility.
- Governed pool fees (ranges e.g. 1–30 bps for stable, 5–100 bps for volatile).
- Integrated oracles
- TWAP on-chain by configurable window (5–30 min) to protect against flash manipulations.
- Export of oracle to KUSD modules/liquidations with safeguards (max delta vs Chainlink/Pyth).

7.3 Fee policy and allocation

- Swap fees
- Collected at the pool level; part LP, part protocol (DAO) governed.
- Redirection of fees (fee-redirect) possible to gKLC gauges.
- Part protocol
- Parameter per pool, capped, allocated to the DAO treasury for security, audits, gasless.
- Reduction/boost
- Fee reductions for addresses/segments (KYB merchants, G2C) if governed and technically feasible via lists.

7.4 gKLC/wKLC : governance of incentives

- wKLC
- KLC staked against wKLC (non-transferable liquid or transferable according to design), harvest on the one hand protocol fees.
- gKLC (vote-escrow)
- KLC Lock to gKLC for duration (e.g. 1 week – 4 years) giving voting rights on 'pools gauges'.
- Emissions/incentives distributed according to gKLC votes; long-term alignment of LPs/manufacturers.
- Gauges and snatches
- Each pool can have a gauge; projects/LP “bribent” gKLC to attract the voting weight.
- Anti-sybil guardrails and concentration limits by entity.
- Sinks and sustainability
- Partial KLC burn (or permanent lock) on protocol fee; slashing for byzantine behaviors of operators if applicable.

7.5 Liquidity management and depth

- Depth objectives
- Stable pairs KUSD/USDC, KUSD/USDT: target depth at 0.5% difference (increments by amounts of 10k/50k/100k KUSD).
- LP Programs
- Targeted incentives by period (8–12 weeks), reviewed based on KUSD peg stability and actual volume.
- Institutional LP
- Dedicated accounts with reporting requirements; semi-passive CL/MM strategies driven by risk.

7.6 Anti-handling, MEV and security

- Protection against flash manipulations
- Mandatory TWAP for sensitive price readings (liquidations/redemptions).
- Local pauses of the pool oracles if deflection > threshold.
- MEV
- Encouragement of fair ordering on canary; sandwich surveillance; public reports.
- Front-running
- Default “deadline + slippage” UX options; batch auctions tested on canary for large institutional blocks.
- Contract security
- Pre-deployment multi-firm audits; active bug bounty; timelocks for critical parameters (A, fees, gauges).
- Prudent governance
- Quorum and deadline to change structural parameters (A, parts protocol, oracles).

7.7 Integrations and composability

- Aggregators
- Intégrations 1inch/0x/Paraswap-like; quote routers exposed in API for PSP.
- Wallets
- Simplified UX swap for major KUSD pairs; clear display of fees and slippage.
- External oracles
- Chainlink/Pyth feed the edge of TWAP; mechanism of deactivation if excessive divergence.
- Governance/Cross-modules
- KUSD liquidations rely on stable pools; gKLC influences the distribution of targeted incentives on peg stability.

7.8 Perimeter compliance

- Sensitive modules (perps/lending)
- Launch only after validated compliance framework (geofencing, KYC opt-in, regional restrictions).
- Lists
- Blocklists for contracts identified as malicious; prevention of snippets from sanctioned addresses.
- Journalisation
- Log requirements for institutional LP partners; accounting export.

7.9 Roadmap produit KalySwap

- V1
- Pools stables KUSD/USDC/USDT; CL pools for KLC/USDC, ETH/KUSD; TWAP on-chain; part minimum protocol.
- V2
- Dynamic third-party fee, hooks (CL policy contracts), unified oracle, fee-redirect to gKLC.
- V3
- RFQ/OTC for institutional blocks; batch periodic anti-MEV auctions; modules compliance “opt-in”.
- Tools
- LP SDK, automatic liquidity management strategies, simulator of return after expenses/impermanent loss.

7.10 Transparency and metrics

- Dashboards
- Volume, TVL per pool, depth at defined thresholds, LP concentration, TWAP price vs external oracles.
- Reports
- Weekly: efficiency of incentives, stability of KUSD peg, deviation events/breaks.
- Published settings
- Amplification A, fees per pool, protocol shares, gauge weights, gKLC emissions.

KalySwap provides robust and measurable liquidity for KUSD/KLC, with optimized stable pools, CL positions for volatile pairs, and incentive governance via gKLC/wKLC that prioritizes stability and depth. The anti-manipulation policy, TWAP oracles and governance discipline aim to make the DEX safe, transparent and institutionalizable, while remaining composable with KUSD and the rest of the ecosystem.

CHAPITRE 8

Tokenomics KLC

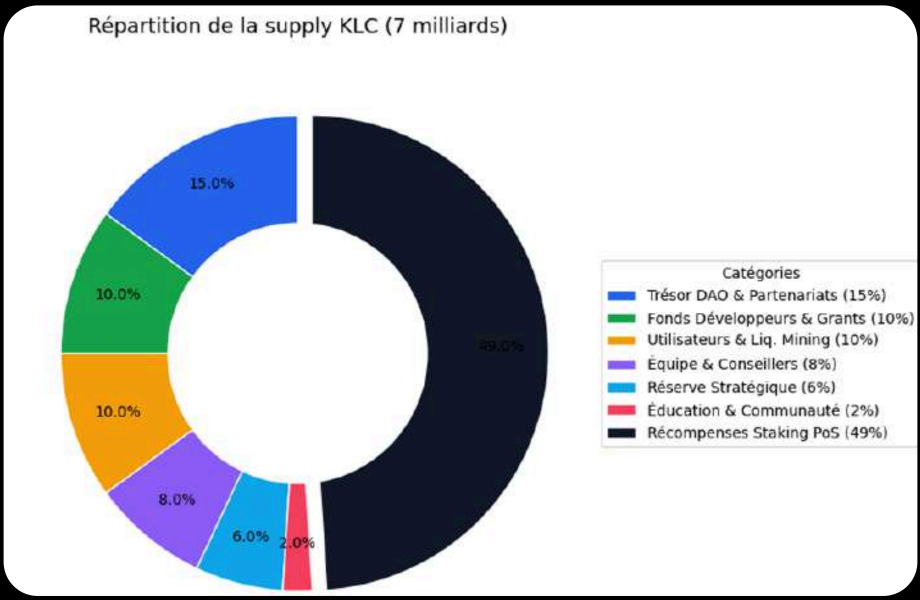
Integrate your supply distribution (7 billion KLC) and specify utilities, emissions/vesting, value capture mechanisms, sinks (burn/locks/slashing) and sustainability. This version replaces and enriches the previous chapter 8.

8.1 Roles and uses of KLC

- Gas and network regulations
- KLC pays the L1 execution fee. For gasless KUSD transfers, the sponsor pays the gas in KLC.
- Staking and security
- KLC is put into play to become a validator or be delegated. The rewards come from the network fee and the 'Staking PoS' pool.
- DAO Governance
- Right of proposal/vote (via gKLC) on DAO budgets, KUSD/KalySwap parameters, gasless policies and network upgrades.
- Capturing ecosystem value
- Governed share of protocol fees (DEX) and KUSD revenues (stability/redemption fees) to the DAO treasury, under legal constraints, which can fund buybacks/burns or gKLC incentives.

8.2 Total supply and distribution

- Total fixed offer: 7,000,000,000 KLC
- Distribution
- 49% Staking PoS: security rewards distributed gradually over 50 years
- 15% DAO Treasury & Partners: strategic alliances, security, audits, integrations, liquidity
- 10% Developer Fund & Grants: dApps, incubators, hackathons, R&D grants
- 10% Users & Liquidity Mining: incentives d’usage (paiements KUSD, LP KalySwap), airdrops ciblés
- 8% Team & Advisors: cliff 12 months, linear vesting 48 months (lock/timelock on-chain)
- 6% Strategic reserve: M&A, regional expansions, exceptional opportunities (use subject to KIP)
- 2% Education & Community: educational programs, workshops, adoption campaigns



8.3 Issuance schedule and policies

- Staking PoS (49% over 50 years)
- Degressive emission curve in five-year increments (e.g. moderate front-load for the first 5–10 years, then decrease).
- Objective: secure 256–1000 validators with competitive APY while converging towards low net inflation.
- DAO/Partners, Dev Fund, Users/Liquidity, Education/Communauté
- Scheduled unlocks and/or multi-signature "vesting" with DAO controls (KIP) for each major disbursement.
- Quarterly usage reports and associated KPIs (integrations, audits, KUSD volume, KalySwap depth).
- Team & Advisors (8%)
- Cliff 12 months then linear vesting over 48 months; no acceleration except in cases of force majeure approved by DAO.
- Strategic reserve (6%)
- Non-circulating by default; case-by-case activation via KIP, timelock, and public justification.

8.4 gKLC and wKLC: incentive alignment

- wKLC
- Liquid staking to capture part of the protocol/DEX flows; short unlocking period (ex: 7 days). Role: flexible participation.
- gKLC (vote-escrow)
- KLC lock (1 week to 4 years) conferring governance rights (votes on KalySwap gauges, DAO budgets, KUSD risk parameters) and possibly LP boosts.
- Governance of incentive issues
- “Gauges” direct the Users & Liquidity Mining distribution according to gKLC votes, with anti-concentration guardrails and eligibility lists.

8.5 Sinks and stabilizers

- Burn
- Partial burn of the base fees (model EIP-1559-like) and possibility of burning on the one hand of 'part protocol' KalySwap if voted.
- Locks
- Encouragement of gKLC long locking (velocity reduction, long term alignment).
- Slashing
- Penalties on validator stakes for double sign, severe downtime, censorship; published policy (thresholds, scales, appeals).
- Buyback (optional, governed)
- Use of protocol/DAO revenue for redemption/burn or incentive reinforcement, under legal constraints and KIP.

8.6 Value flows between modules

- KalySwap Trésor DAO
- Part protocol of swap fees allocated to the treasury for: security/audits, gasless, grants, and possibly KLC buybacks/burns.
- KUSD Trésor DAO
- Stability/redeem fees (Core/Plus) assigned to: oracles, risk buffers, audits, operations; buyback/burn possible if deemed appropriate by the DAO.
- Publication
- Quarterly budgets, limits by category (e.g., min 30% safety/audits; max X% buyback), implementation reports.

8.7 Economic security and attack costs

- Hypotheses
- Target staking rate: 40–70% of circulating supply; set of 256–1000 validators under HotStuff.
- Attack cost 1/3
- Capital barrier + slashing risk + destruction of reputational value disincentive alignment.
- Adjusting rewards
- Adjustable PoS issuance parameters (KIP + timelock) to preserve finality/SLO when network fees drop.

8.8 Sustainability scenarios

- Base
- Moderate network charges; increasing KUSD/KalySwap usage; degressive PoS emissions; regular burns.
- Bear
- Low volume: maintaining APY via scheduled PoS issuance; reduction of LM incentives; priority to security/operations expenses.
- Bull
- High volumes: net inflation reduction (up to 0–1%); increase in burns; constitution of countercyclical reserves.

8.9 Reporting and transparency

- Public dashboards
- PoS emissions achieved vs plan 50 years; actual distribution; burn tracker; gKLC/wKLC TVL; treasure flow.
- Periodic reports
- Monthly: issues, burns, DAO expenses by category, impact KPIs (KalySwap depth, KUSD volumes).
- Quarterly: allocation audit of pockets (DAO/Partners, Dev Fund, Users/LM, Education, Reserve).
- Addresses/contracts
- Vesting/treasury timelocks, gKLC/wKLC, gauges and parts protocol documented in Appendices.

8.10 Legal framework and good practices

- Separation Foundation/DAO/KALYSSI
- Flow allocation and compliant communications; avoid any promise of return.
- Disclosures
- Risks: token volatility, governance, liquidity, technical execution; no guarantee of performance.

KLC, with a fixed offer of 7 billion, is distributed to sustainably secure the network (49% PoS over 50 years), finance the ecosystem (DAO/Partners, Dev Fund, Users/LM, Education), align the team (strict vesting) and preserve a governed strategic reserve. The gKLC/wKLC mechanisms, sinks (burn/locks/slashing) and the degressive issuance policy aim at a long-term balance between security, adoption and budgetary discipline, under an on-chain transparency and governance framework.

CHAPITRE 9

Operational Governance : Foundation + DAO

Define precisely the separation of roles between the KalyChain Foundation, the DAO and KALYSSI, formalise the proposal/voting process (KIP), detail the specialised committees (risk, security, listings) and structure the Transparency Hub as an anchor point for public trust.

9.1 Guiding principles of governance

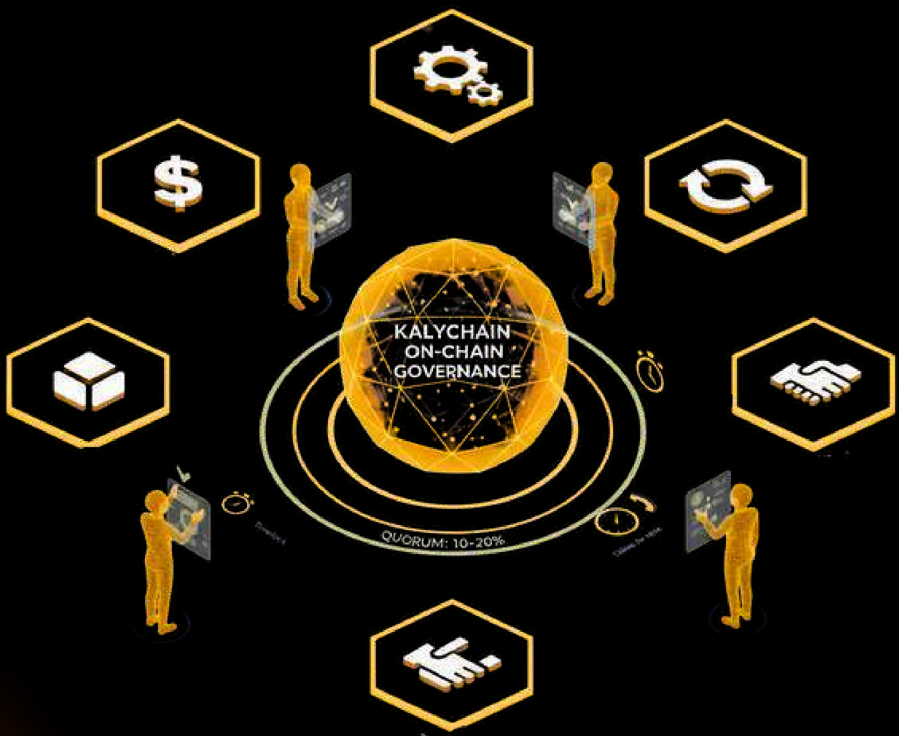
- Separation of powers
- Foundation: legal execution, operations, compliance, institutional representation.
- DAO: strategic decisions, budgets, protocol parameters, product orientation.
- KALYSSI: separate commercial entity (CASP licenses), without direct control over the protocol.
- Radical transparency
- All votes, budgets, parameters and incidents published on the Transparency Hub.
- Documented treasury addresses, governance contracts, timelocks and multisigs.
- Progressivity and security
- Mandatory timelocks for critical changes (consensus, KUSD, bridges).
- Guardian multisig with limited mandate and ex post control by the DAO.

9.2 KalyChain Foundation: role and scope

- Legal status
- Non-profit foundation, domiciled in Panama, with public charter and independent board of directors.
- Mandates
- Execute the decisions of the DAO (budgets, listings, partnerships).
- Manage institutional relations (regulators, banks, PSP, exchanges).
- Operate critical infrastructure (public RPC, indexing, gasless relays).
- Coordinate audits, bug bounty, and incident response.
- Limits
- No discretion on protocol parameters (KUSD, KalySwap, gasless) without DAO vote.
- Public quarterly reporting: budgets executed, partnerships signed, operational KPIs.
- Board of Directors
- Independent members (blockchain experts, finance, compliance, tech).
- Mandates renewable by DAO vote; conflicts of interest declared publicly.

9.3 DAO KalyChain: structure and power

- Composition
- Holders of gKLC (vote-escrow) with weight proportional to the amount and duration of lock.
- Possible delegation to representatives (delegates) with transparency on-chain.
- Decision areas
- Budgets: quarterly allocation of DAO Treasury, Dev Fund, Users/LM, Education, Strategic Reserve.
- Protocol parameters: KUSD (LTV, fees, caps), KalySwap (A, fees, gauges), gasless (quotas, budgets, eligibility).
- Network upgrades: HotStuff activation, hard forks, consensus changes.
- Listings and partnerships: validation of criteria, approval of tier 1/tier 2 exchanges.
- Committees: appointment/dismissal of members of specialised committees.
- Voting process
- Quorum: minimum threshold of gKLC participant (ex: 10–20% of the locked offer).
- Majority: simple (>50%) or qualified (>66%) depending on the criticality.
- Duration: voting period 5–14 days depending on urgency; timelock post-vote 2–14 days depending on the impact.



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- Duration: voting period 5–14 days depending on urgency; timelock post-vote 2–14 days depending on the impact.

9.4 Processus KIP (KalyChain Improvement Proposal)

- Phases
- Discussion (Forum): informal proposal, community debate, technical feedback.
- KIP Draft: formal drafting (context, specification, impacts, risks, costs).
- Review: concerned committees (risk, security, tech) issue an advisory opinion.
- Vote on-chain: snapshot gKLC, voting period, quorum and majority required.
- Timelock: lead time (2–14 days depending on criticality).
- Execution: Foundation or autonomous contracts apply the decision.
- Post-mortem: impact report published 30–90 days after.
- Templates and standards
- KIP-0: Metagovernance (process itself).
- KIP-1xx: Economic parameters (KUSD, KalySwap, tokenomics).
- KIP-2xx: Technique (consensus, upgrades, bridges).
- KIP-3xx: Budgets and Allocations.
- KIP-4xx: Partnerships and listings.
- Tools
- Official forum (Discourse/Commonwealth), Snapshot/Tally for on-chain votes, GitHub for technical specifications.

9.5 Specialised committees

- Risk Committee
- Mandate: propose and monitor KUSD risk parameters (LTV, caps, oracles), quarterly stress tests, alerts on concentration/collat.
- Composition: 5–7 members (challenge experts, risk managers, auditors), appointed by DAO, renewable 12-month terms.
- Deliverables: quarterly public reports, KIP parameter proposals, risk dashboards.
- Listings & Liquidity Committee
- Mandate: evaluate CEX/DEX listing requests according to technical/qualitative criteria (audits, depth, stability, integrations); coordinate market makers.
- Composition: 5 members (market experts, MM, compliance), appointed by DAO.
- Deliverables: public evaluation grids, due diligence reports (anonymized if NDA), recommendations to the DAO.

- Payments & Gasless Committee
- Mandate: propose gasless policies (quotas, budgets, eligibility), follow adoption KPI, coordinate PSP/wallets integrations.
- Composition: 5 members (payment experts, UX, compliance), appointed by DAO.
- Deliverables: monthly gasless usage reports, adjustment proposals, impact studies.
- Transparency of the committees
- Public meetings (or public reports); declared conflicts of interest; remuneration in governed KLC (DAO budgets).

9.6 Separation with KALYSSI

- Role of KALYSSI
- Commercial entity holding CASP licenses (service provider on digital assets).
- Provides KYC/KYB services, banking integrations, Travel Rule compliance for institutional flows.
- Operates as a protocol partner, with no control over governance or settings.
- Perimeter
- KALYSSI does not vote on the DAO (except for personal ownership of gKLC by its shareholders, declared).
- Commercial contracts between KALYSSI and users/institutions are distinct from the protocol.
- KALYSSI audits and reports published separately (compliance, licenses).
- Transparency
- Documented legal/financial links between Foundation and KALYSSI; no undeclared cross-cash flows.

9.7 Guardian Multisig and emergency powers

- Composition
- 5–9 signatories (Foundation members, core devs, elected DAO representatives), threshold 3/5 or 5/9.
- Limited powers
- Emergency break: bridges, sponsor gasless, mint KUSD Plus in case of critical incident (oracle failure, exploit detected).
- No power of confiscation, arbitrary censorship or modification of parameters without urgency.
- Ex post control
- Any pause activation must be publicly justified within 24 hours.
- Complete incident report within 7 days.
- Vote DAO of ratification or revocation within 14 days; if revoked, the guardians are replaced.

9.8 Transparency Hub: trusted infrastructure

- Content
- Governance: history of KIP, votes, results, active timelocks, composition of committees.
- Finances: treasury addresses (DAO, Foundation, vesting), quarterly budgets, expenses incurred, protocol revenue.
- KUSD: Core/Plus reserves in real time, strike ratios, caps, active oracles, redemption/liquidation history.
- Network: SLOs (availability, purpose, TPS), incidents, post-mortems, upgrade schedule.
- Security: published audits (links), bug bounty stats, contract addresses, ABIs, binary checksums.
- Gasless: cumulative costs, volumes, aggregate beneficiaries, active policies.
- Technologies
- Official Subgraphs The Graph, Dune/Flipside dashboards, dedicated website with public APIs.
- IPFS/Arweave archiving for critical reports (audits, postmortem, votes).
- Update
- Real time for on-chain metrics; weekly for committee reports; quarterly for financial audits Foundation.

9.9 Delegation and participation

- Delegation of votes
- gKLC holders can delegate their voting power to representatives (delegates) with instant revocation.
- Delegates publish their positions and conflicts of interest; classification by delegated weight.
- Incentives for participation
- Symbolic rewards (KLC) for active participation (votes, proposals adopted).
- Governance education programmes (workshops, simulations).

9.10 Evolution of governance

- Gradual decentralization
- Phase 1 (2024–2025): Foundation executes, DAO validates major decisions, advisory committees.
- Phase 2 (2026–2027): DAO assumes budgets and parameters, Foundation becomes pure execution, binding committees.
- Phase 3 (2028+): maximum automation (autonomous contracts), reduced foundation, mature and self-sufficient DAO.
- Metagovernance
- The KIP process itself is governed: quorums, majorities, timelocks, composition of committees can be modified by KIP-0xx.

KalyChain governance is based on a clear separation Foundation (legal/operational execution)/ DAO (strategic decisions and parameters)/ KALYSSI (regulated commercial services), orchestrated by a transparent KIP process, specialized committees (risk, security, listings, gasless) and a comprehensive Transparency Hub. Safeguards (timelocks, limited multisig guardian, ex post control) and progressive decentralization aim to reconcile agility, security and institutional legitimacy.

CHAPITRE 10

Security and Resilience

Map KalyChain's attack surface (L1, KUSD, KalySwap, governance), define security requirements by design (audit, fuzzing, formal verification), OpSec practices (keys, HSM, access), dependency management (oracles, bridges, relayers), and the incident/rollback playbooks. This chapter concludes with security SLO/SLAs, public metrics and rolling audit schedule.

10.1 Attack surface and threat hypotheses

- L1/Consensus/Clients
- Client implementations (HotStuff migration), mempool DoS, reorgs, censorship, double-sign.
- P2P networking exploits, liveness/safety consensus, fragmentation of the validator set.
- Smart contracts
- KUSD (mint/redeem, collat Core/Plus), KalySwap (AMM stables, CL/MM, gauges), gKLC/wKLC.
- Classes de bugs: reentrancy, overflow/underflow, price manipulation via oracles/pools, griefing sur refunds.
- Governance
- gKLC takeover (malicious vote-buying), committee capture, configuration errors via KIP.
- External dependencies
- Oracles (Chainlink/Pyth), bridges, indexers, RPC, gasless relays.
- Operations
- Multisig/foundation keys, identity leaks, ransomware, supply chain (libraries, CI/CD).

Threat model: rational attacker, significant financial resources, MEV capability, access to flash loans, partial collusion of validators, legal attacks (takedown).

10.2 L1 Protocol Security

- Consensus and clients
- Migration to HotStuff with fast finality; proven safety properties; tolerance $f < n/3$.
- Testnets canary/ staging with chaos engineering (network latency, partitions, churn of validators).
- Anti-DoS and mempool
- Adaptive pricing, limits by peer, gasless tx bucketisation, block quotas (see Chap. 6).
- Network hardening
- Peer scoring, rate limiting, banning temporaire, protection contre eclipse attacks.
- Client/infra diversity
- Two independent implementations if possible; diversity of OS, cloud/on-prem providers; version monitoring.

10.3 Security of smart contracts

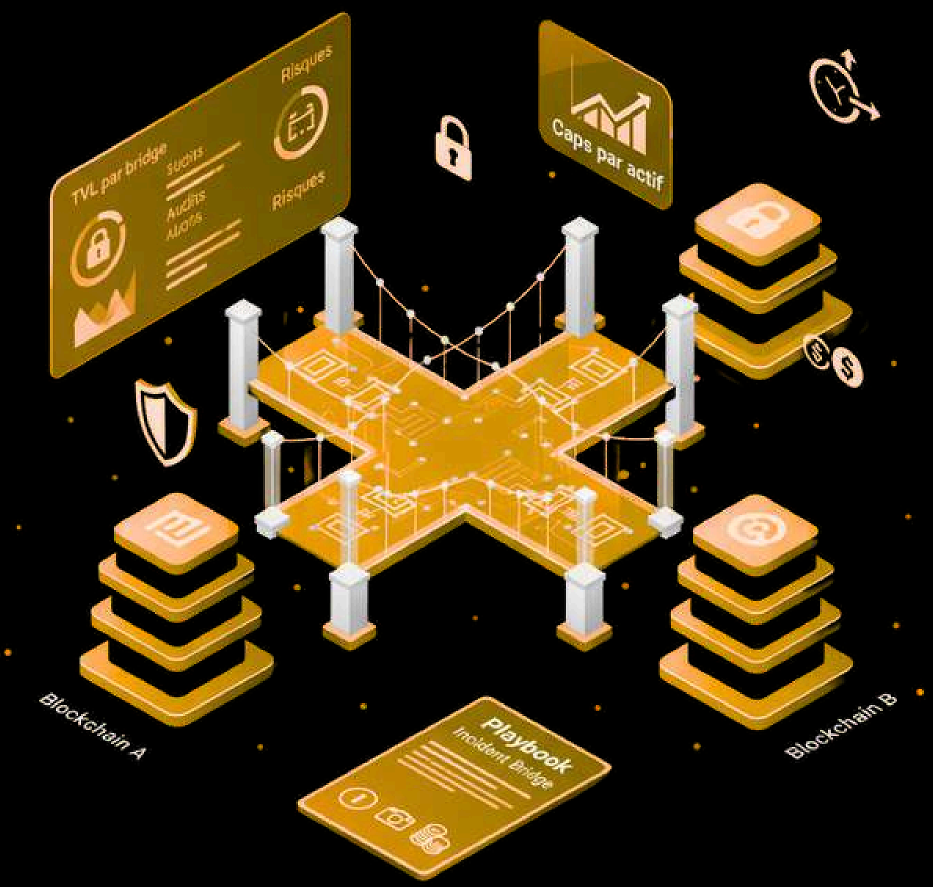
- Standards and patterns
- OpenZeppelin, checks-effects-interactions, pull over push pour transferts, pausable guard rails limité.
- Audits multi-firmes
- 2–3 cabinets before production for KUSD, KalySwap, gKLC/wKLC, gauges.
- Re-audit at each major upgrade; full publication of the reports.
- Fuzzing and properties
- Fuzzing differential et invariant testing (Foundry/Medusa/Slither).
- Targeted formal verification for critical invariants:
- KUSD: retention of collateral, LTV terminals, impossibility of uncollateralized mint.
- Stable MAs: invariant of conservation, slippage limits, TWAP consistency.
- gKLC: conservation of vote-power, lock/timeline, impossibility of ungoverned dilution.
- Bug bounty
- Public program, tiered bonuses, fast-track triage, responsible disclosure; bonus for reproducible PoC.

10.4 Oracles, price and anti-manipulation

- Internal TWAP + external oracles
- TWAP 5–30 min for sensitive decisions (liquidations/redemptions).
- Cross-check with Chainlink/Pyth; divergence thresholds and "circuit breakers" temporary.
- Multiple sources
- Multi-oracles aggregation when possible; failover/pauses if 2+ sources diverge strongly.
- Flash attacks
- Instant spot readings rejections; minimum delays; minimum depth required for price-read.

10.5 Bridges and interoperability

- Policy bridges
- Prefer bridges “light client”/ verifiable; avoid the single centralized custody.
- Caps per bridged asset; timelocks of increase; risk dashboards (TVL by bridge, audits).
- Playbook incident bridge
- Pause of mint/redemptions of compromised assets; snapshots; recapitalization/bailout plan governed.



10.6 OpSec: keys, access, CI/CD

- Key management
- Multisigs 3/5 or 5/9 for treasures/guardians; distinct thresholds for routine vs critical operations.
- HSM/Cloud KMS for signatures; planned rotation; off-site storage for shards.
- Access and role separations
- Strict RBAC, mandatory MFA, access “just-in-time” and immutable logs.
- Séparation dev/ops/audit; four-eyes for deployments.
- Supply chain
- Pinning dependencies, SLSA level 2+, build signatures, SCA/DAST/IAST scans.
- CI/CD
- Isolated environments, canary deployments, "limited and governed" kill switch; published artifacts (hash, checksum).

10.7 Operational resilience

- Redundancy
- RPC and multi-region indexers; encrypted backups; disaster recovery (RTO < 4h, RPO < 1h).
- Observability
- Metrics, logs, traces; alerting p50/p95 latency, error rate, mempool backlog, purpose.
- Chaos tests
- Failure scenarios of cloud regions, oracles intrusion, mempool overload, network partition between validators.

10.8 Incident playbooks and rollback

- Classification of incidents
- Severity SEV-1 to SEV-4; criteria (loss of assets, network shutdown, data corruption, degradation of SLO).
- Answer
- War room (Security Committee + core devs + Foundation Ops); public communications in < 60 min.
- Actions: targeted pause (AMM/KUSD/gasless), oracle deactivation, mint cap, quota reduction.
- Rollback/patch
- Preference for hotfix forward; hard fork/rollback only if required for balance integrity.
- Documented procedure: snapshot, migration scripts, emergency votes, post-fork checks.
- Post-mortem
- Posted under 7 days: timeline, root cause, impact, preventive measures, follow-up actions.

10.9 SLO/SLA security and public metrics

- SLO network
- Purpose: median 2 blocks; p95 5 blocks. Availability consensus 99.95%.
- SLO DeFi
- Precision of the oracle: divergence TWAP vs external 0.5% in regime; pause time < 10 min if > threshold.
- SLO operations
- RTO < 4h, RPO < 1h; MTTR SEV-1 < 24h.
- Published metrics
- Incident rate per quarter, audit coverage (contracts, lines of code), success rate fuzzing, bug bounty participation, average KIP application time critical.

10.10 Continuous audit schedule

- Before mainnet
- 2–3 independent audits for KUSD, KalySwap, gKLC/wKLC; client audit/consensus; pen-test infra Foundation.
- Post mainnet (year 1)
- Semi-annual audit of critical modules; continuous fuzzing; quarterly external challenge (CTF).
- Year 2+
- Program “continuous assurance”: incremental coverage, formal verification of added invariants, annual red-team.

KalyChain’s security combines in-depth defenses (robust consensus, multi-firm audits, fuzzing, targeted formal verification), strict OpSec practices (HSM, multisigs, secure CI/CD), prudent dependency management (oracles, bridges) and clear incident playbooks. Transparency via public SLO/Metrics and a continuous audit schedule strengthens resilience and credibility with users and institutions.

CHAPITRE 11

Compliance, legal and partnerships institutional

Define the KYC/KYB and Travel Rule frameworks, the Foundation/DAO/KALYSSI segregation, the regulatory requirements by perimeter (stablecoin, DEX, staking), and the structuring of partnerships (banks, PSP, card issuers, oracles, custodians). This section specifies the internal controls, traceability and contractual guarantees expected by the institutions.

11.1 Compliance framework: principles and scope

- Approach “by design”
- Modularity: opt-in compliance layers (KYC/KYB, sanctions lists, flow rules) can be activated by segment (institutions, PSP).
- Data minimisation: proportionate collection, limited retention, encryption and restricted access.
- Regulated perimeters
- Stablecoin KUSD (Core/Plus): AML/CFT requirements, certificates of reserves, oracles of price, transparency of collaterals.
- KalySwap DEX: listings policy, blocklists, manipulation monitoring, risk disclosure.
- Staking/gKLC: careful communication (no promise of return), compliance with applicable securities laws.
- Commercial entities (KALYSSI): partner CASP/PSAN/EMI licenses according to the jurisdictions served.

11.2 KYC/KYB, Sanctions and Travel Rule

- KYC/KYB
- Process via KALYSSI and/or certified partners (capture document, liveness, PEP/sanction screening).
- Segmentation: individuals (standard KYC), companies/PSPs (reinforced KYB), NGOs/governments (dedicated flows).
- Sanctions/Blocklists
- Continuous screening (OFAC, EU, UN) and internal lists of risky smart contracts; application side relayers/PSP and deny-lists on-chain minimales.
- Travel Rule
- Intégration TRISA/Travel Rule Universal Solution via partenaires VASP; exchanges of inter-VASP encrypted data; triggering beyond regulatory thresholds by region.

11.3 Segregation Foundation/ DAO/ KALYSSI

- Foundation
- Operational and legal execution; holds and publishes contracts, treasury addresses, audit reports; does not hold unilateral policy powers.
- DAO
- Governance of parameters (KUSD, KalySwap, gasless) and budgets; public decisions via KIP and timelocks.
- KALYSSI
- Commercial Entity (CASP) for KYC/KYB, fiat rails, cards and PSP; no discretionary influence on the protocol. Separate service contracts, annual audits.

11.4 Bonds for KUSD (Core/Plus)

- Reservations and certificates
- Periodic certificates by third-party firm; real-time reporting of collaterals (Core/Plus) on the Transparency Hub.
- Risk governance
- Parameters (LTV, caps, haircuts, fees) proposed by the Risk Committee and voted by the DAO; “circuit breakers” in case of stress.
- Legal
- Terms and disclosures: nature of the stablecoin, market risks, redemption procedures; compliance by jurisdiction of collateral (securities/currency).

11.5 Listing policies and markets

- Listing DEX
- Criteria: valid audits, sufficient liquidity, absence of sanctions, clear documentation; possibility of delisting via KIP in case of risk.
- CEX and route-to-market
- Technical records and compliance; market making framed (transparent liquidity contracts); non-misleading communication.

11.6 Partnerships PSP, banks, cards, custodians

- Banks and EMI
- Settlement accounts, multi-currency on/off-ramps, merchant encashment programs; SLAs (D+1), reconciliations, chargebacks.
- PSP/Acquirers
- Plugins (Woo/Shopify-like), accounting, refunds; co-sponsoring gasless; anti-fraud reporting.
- Card networks
- Prepaid/virtual card programs (where legal); fiat KUSD conversion in real time; conformance chargeback/KYC.
- Custodians
- Guarding of KUSD Core reserves, institutional staking, HSM; SOC 2/ISO 27001 certificates; cold/warm/hot segregation.

11.7 Oracles, auditors and critical suppliers

- Oracles
- Multi-suppliers (Chainlink/Pyth) with SLAs, monitoring and contractual recourse; divergence thresholds triggering pauses.
- Auditors
- Code audits (smart contracts/clients), certificates of reserves, infra pen-tests; publication and tracking of remediations.
- Independence
- Periodic rotation of cabinets; no dependence on a single supplier for critical components.

11.8 Internal controls, data protection and reporting

- Internal controls
- Segregation of duties, RBAC, immutable newspapers, quarterly journals; customer acceptance policies (KYC risk scoring).
- Data protection
- Encryption at rest/in transit, minimization, DPA, data localization according to local laws (ex: GDPR).
- Reporting
- Foundation Quarterly Reports (finance, incidents, compliance); monthly DAO reports (budgets, parameters); institutions portal with metrics and SLA.

11.9 Legal mapping and geographies

- Target countries
- Priority: LATAM, Africa, emerging Asia for payments/remittances; Europe/NA via licensed partners.
- Mapping
- Analysis by jurisdiction: VASP/CASP, stablecoin rules, sanctions, data transfer; go-to-market matrix by segment (retail, SME, institution).
- Georestrictions
- Access blocking for jurisdictions under strong sanctions; local compliance via KALYSSI/partners.

11.10 Contractual clauses and SLAs

- SLAs partners
- Availability > 99.9% for critical APIs; latency p95 < 300 ms; settlement deadlines; incident resumption (RTO/RPO).
- Compliance clauses
- Audit rights, AML/CFT obligations, TR clauses, confidentiality and security (SOC 2/ISO), termination for cause (sanctions/violations).
- Revisions
- Annual reviews, adjustments according to new regulations; joint partner governance committees.

KalyChain aligns with pragmatic and modular compliance: KYC/KYB/Travel Rule via certified partners, transparent on-chain governance, regular certificates of KUSD reserves and robust listing policies. Foundation/DAO/KALYSSI separation, clear SLAs with banks/PSP/oracles/custodians, and strict internal controls create a reliable framework for institutional adoption while preserving the openness and composability of the protocol.

CHAPITRE 12

10 years roadmap and indicators of success

Consolidate a 10-year trajectory, without reference to capitalization or ranking objectives, with technical/product milestones, adoption targets by segments (retail, SMEs, institutions), and operational, economic and governance health indicators. The roadmap is governable via KIP and subject to continuous adaptation.

12.1 Planning principles

- Division into governable phases: each major milestone passes through KIP (specifications, budget, risks, SLO).
- Robustness before complexity: security, perimeter compliance, UX, observability prioritized compared to products "long tail".
- Measurement and transparency: objectives expressed in SLO/SLA, adoption, liquidity depth, stability of KUSD, not in speculative metrics.
- Iteration and canary: progressive production implementation (canary/staging), feature flags, documented rollbacks.

12.2 Roadmap by phases

Phase I — Foundations (Years 1–2)

L1 and consensus

QBFT stabilization migration HotStuff (fast finalization), client diversity, SLO monitoring.

Payments and KUSD

KUSD Core/Plus launch, multi-vendor oracles, secure liquidations, recurring reserve attestations.

Gasless V1 program (KUSD transfers), quotas/anti-abuse, public dashboards.

basic challenge

KalySwap V1 (pools stables KUSD/USDC/USDT, CL/MM principales), TWAP, part protocole minimale.

Governance and security

gKLC/wKLC V1, KIP process, committees (risk, security, listings, gasless).

Multi-firm audits, bug bounty, network chaos tests.

Partnerships

2–3 PSP/acquirers, 1–2 banks/EMI, 2 oracles, 1–2 custodians. Intégrations e-commerce (plugins).

Target adoption

50–150 merchants/PSPs integrated; 50–200k active wallets; 10–25m$ TVL pools stables; SLO finality p95 5 blocks.

Phase II — Scale and Resilience (Years 3–4)

L1 and infra

Optimizations mempool/fee-market, gasless dynamic quotas; Multi-region RPC/indexing; SLO availability 99.95%.

KUSD and payments

KUSD V2: enhanced risk buffers, improved liquidations, real-time reporting of reserves; G2C/ONG channels.

Gasless V2: co-sponsoring PSP/banks, batch payouts, segmented rules (retail/SME/institution).

extended challenge

KalySwap V2: dynamic third-party fee, policy hooks, unified oracle, fee-redirect to gKLC.

Cautious subsidiary markets (stable high-quality collateral lending; compliant scope).

Governance

Structured gKLC delegation, continuous audits, committee rotation, reports SLAs.

Partnerships

5–8 Multi-Region PSP/Acquirers; 3–5 banks/EMI; institutional custodians; major swap aggregators.

Target adoption

500–1 500 merchants; 300–800k active wallets; 100–$ 300m cumulative depth on stable KUSD pairs; uptime 99.95%.

Phase III — Interoperability and regional (Years 5–6)

Interop and bridges

Selection of verifiable bridges; caps per asset; monitoring risk bridge; major L2/L1 connections.

Payments

Card programs (or legal via partners), regional remittances; extended fiat rails (LATAM, Africa, Asia).

DeFi pro

RFQ/OTC for institutional blocks; batch auctions anti-MEV; collateralized credit lines for SMEs under KYB.

Compliance

Travel Rule end-to-end via VASP partners; fine georestrictions; ISO/SOC certifications for Foundation.

Target adoption

3,000–8,000 merchants; 1.0–2.5M active wallets; >$500m depth KUSD pairs; latency p95 swaps < 2 s on RPC partners.

Phase IV—Automation and Maturity (Years 7–8)

Automation governance

Self-regulated parameters under DAO custody (control bands for A, fees, gasless quotas); on-chain executions without Foundation intervention excluding exceptions.

Treasure and sustainability

Countercyclical policies (reserves/stability fees); buyback/burn governed; net inflation scenario 0–1% if fees are sufficient.

Products

Tokenized financial services (factoring/escrow/parametric insurance) via compliant partners; KYC opt-in modules by flow.

Target adoption

8,000–20,000 merchants; 3–6M active wallets; >1.5–3.0B$ stable depth KUSD; SEV-1 incidents 1/year.

Phase V — Standardization and Sustainability (Years 9–10)

Standardization

Open Specifications (KIP) published and adopted by partners; certification of validators and relayers.

Resilience

Full client diversity; annual Red Team exercises; RTO < 2h, RPO < 30 min; MTTR SEV-1 < 12h.

Ecosystem

Expanded university/education programmes; regional hubs of integrators; security/MEV search bounties.

Target adoption

20,000–50,000 merchants; 6–12M active wallets; >3–6B$ stable depth; KUSD stability (average spread < 10 bps in terms).

12.3 Success indicators (KPI) by domain

- Network and performance
- Median/Blocks Purpose; p95 latency; uptime consensus/RPC; backlog mempool; diversity validators; concentration stake.
- Security
- Audit coverage; vulnerabilities fixed; participation bug bounty; incidents by severity; malicious MEV detection.
- KUSD
- Core/Plus bond ratio; peg stability (TWAP vs external); volumes mint/redeem; successful liquidations; oracles pause time.
- challenge/liquidity
- Depth KUSD stable pairs at thresholds 10k/50k/100k; slippage observed; protocol shares collected; efficiency of incentives.
- Payments/gasless
- Tx gasless/tx totales KUSD; average cost/sponsored tx; active beneficiaries; rate of fraud/abuse; repayment terms relayers.
- Adoption
- Active merchants; volume processe; monthly active wallets; cohort retention; partner integrations (PSP, banks, aggregators).
- Governance
    - gKLC participation rate; average execution time KIP; voting concentration; transparency budgets; compliance reporting committees.
- Operations/Compliance
    - Respected SLA partners; deadlines KYC/KYB; successful Foundation audits (ISO/SOC); data incidents.

12.4 Guardrails and phase transition criteria

- Guardrails
- High frequencies for critical modifications; timelocks; spending limits by category; risk caps (KUSD, bridges).
- Passage criteria
- Phase I II: SLO finality p95 5 blocks over 90 days; past major audits; stable depth 25–50m$; 500 merchants.
- Phase II III: uptime 99.95% over 180 days; operational Travel Rule systems; stable depth $250m; 3k merchants.
- Phase III IV: SEV-1 incidents 2/year; customer diversity; >1B$ stable depth; partial automated governance.
- Phase IV V: MTTR SEV-1 < 12h over 12 months; standardization adopted by partners; >3B$ stable depth; 20k merchants.

12.5 Transparency and revisions

- Pace of review
- Quarterly: progress status roadmap, budgets, KPIs; adjustments via KIP.
- Annual: strategic review 10 years rolling; update of the passage criteria.
- Tools
- Transparency Hub: public dashboards, API exports, IPFS/Arweave archives, committee reports.

This decadal roadmap prioritizes security, payment UX, KUSD stability, liquidity depth and transparent governance. It sets measurable milestones and clear safeguards, while remaining governable and adaptable by KIP. The objective is not a 'market place', but a reliable and compliant infrastructure that sustainably serves trade, remittances and on-chain finance.

Conclusion

KalyChain proposes a pragmatic and coherent response to the limits that still hinder the adoption of on-chain finance by citizens, SMEs, and public institutions. By combining a secure L1 EVM, a fast finality under HotStuff, an overcollateralized and auditable stablecoin (KUSD), governed gasless transfers, and a hybrid Foundation + DAO governance anchored in transparency, the architecture emphasizes reliability, UX and compliance where most existing rails prioritize pure performance or speculation.

This white paper details the guardrails that condition the long-term credibility of the protocol: public SLO/SLA, multi-firm audits, verification of critical invariants, oracles and bridges under caps and pause policy, timelocks, multisig guardian with limited powers, specialized committees and a comprehensive Transparency Hub. The KLC tokenomics, designed on a degressive issuance trajectory and alignment mechanisms (gKLC/wKLC, burns/locks/slashing), aims at a balance between economic security of the network, liquidity incentives and budgetary discipline.

The ten-year roadmap is built around measurable objectives of KUSD stability, depth of stable pairs, p95 purpose, merchant/wallet adoption, uptime and transition criteria with safeguards. It prioritizes responsible scalability: first robustness (security, perimeter compliance, observability), then scale (PSP/bank partnerships, G2C, remittances), finally automation of governance and standardization.

We do not pursue either the race for ranking or speculative capitalization. The ambition is to establish reliable, interoperable and auditable payment and institutional finance channels capable of supporting real flows and constrained by regulations. If the community validates this trajectory through the KIP process and operational guardrails are respected—KalyChain can become a reference standard for on-chain payments and SME treasury, and a credible bridge between traditional finance and challenge.
Call for input: we invite developers, auditors, PSP, banks, public institutions and communities to participate in canary tests, architecture reviews, audits, and the definition of gasless policies and KUSD risk parameters. Trust is built by proof: audited code, public metrics, documented incidents, reasoned votes. It is on this basis, and not on promises, that KalyChain intends to gain legitimacy and serve the real economy sustainably.

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